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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
IV Semester B. E. Examinations Oct/Nov - 2022
Aerospace Engineering
AEROSPACE STRUCTURES

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

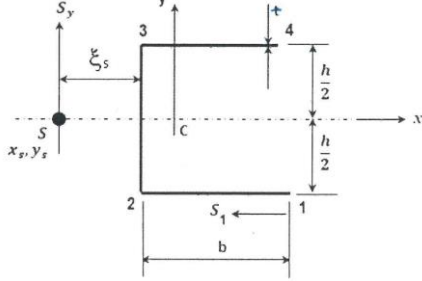
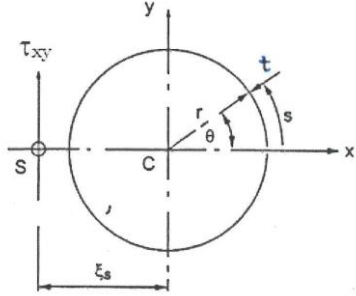
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	The structure of an aircraft is required to support two distinct classes of load which are termed as _____ and _____.	02
	1.2	Aircraft designed for high altitude flight must withstand _____ pressure.	01
	1.3	In aircraft structures longitudinal stiffeners are used to divide the skin into narrow panels thereby increasing the _____ of the skin.	01
	1.4	What is V-n diagram?	01
	1.5	What is shear center in open section beams?	01
	1.6	The safe-life of an aircraft is predicted based on _____.	01
	1.7	Fail Safe designs are designs that incorporate various techniques to mitigate losses due to _____ failures.	01
	1.8	A torsion box is used in an aircraft wing to provide _____.	01
	1.9	Failures of landing gears are mainly due to _____.	01
	1.10	_____ welding process is used for welding of sheet metals in aircraft industries.	01
	1.11	The theory of the torsion of closed section beams is known as the _____ theory.	01
	1.12	For a given material under buckling, the critical stress increases as the _____ decreases.	01
	1.13	The combined action of skin and stringers is to resist _____ & _____ loads.	02
	1.14	High-frequency stress fluctuations, due to vibrations excited by jet or propeller noise, cause _____ fatigue.	01
	1.15	Why aircraft structures are riveted instead of welding? Comment.	02
	1.16	Damage Tolerant design philosophy is used by aircraft designers to minimize the effects of _____ and _____.	02

PART-B

2	a	With the aid of a neat flight envelope diagram explain the performance of a flight.	10
	b	Discuss different type of 'loads' acting on aircraft structural components.	06

3 a b	Explain in detail the influence of shear center on Aero elastic instability. Calculate the position of the shear center of the thin-walled channel section shown in Fig. 3.b. The thickness 't' of the walls is constant.  Fig. 3.b	06 10
OR		
4 a b	Compute the shear flow distribution in the thin-walled open section shown in Fig. 4.a produced by a vertical shear load, τ_{xy}, acting through its shear center. Also sketch the shear flow distribution.  Fig. 4.a Explain in detail the shear of closed section beams.	10 06
5 a b	What is critical load? Show that at critical value of the load the structure will be in equilibrium and independent of the magnitude of angle 'θ'. Explain with equations Gerard's method to evaluate crippling stresses. OR Obtain an expression for critical load for a column subjected to eccentric axial loads. Calculate the critical load of a column which is made of a bar, circular in cross section and 5m long, which is pinned at both ends. The same bar when simply supported gives a mid-span deflection of 10mm with a central concentrated load of 10N.	10 06 12 04
7 a b	Discuss the 'safety factor' in the design of aircraft structures. Distinguish between 'fail safe design' and 'damage tolerance design' in aircraft structures.	08 08
8 a b c	With aid of neat diagram, briefly explain the Lug strength analysis under transverse loading condition. Explain the four failure modes of riveted connections with relevant equations. List any two advantages of welded connections over riveted connections.	06 08 02